

Snowflake Cortex Governance Health

Check Skill by Ishrak Wasif Uday

AI-assisted risk evaluation workflow for Snowflake data governance

Project summary: I designed and tested a read-only Cortex skill that inspects Snowflake metadata, identifies governance gaps, and converts technical findings into a concise risk report with severity levels and recommended next steps.

Area	Details
Project type	Enterprise AI workflow, data governance, metadata-driven risk analysis
Core artifact	SKILL.md for a Snowflake governance health-check workflow
Tools	Snowflake, Cortex Code CLI, SQL, ACCOUNT_USAGE, INFORMATION_SCHEMA, Markdown
Primary users	Data governance, platform, security, and analytics teams
Design principles	Read-only, explainable, metadata-driven, scope-aware, and safe by default

1. Project Overview

The goal of this project was to make Snowflake governance reviews faster and easier to interpret. Instead of manually checking tables, tags, grants, masking coverage, and row-level controls across multiple metadata views, the skill guides a user through a repeatable governance health check and returns a structured report.

The workflow is designed for practical enterprise use: it does not modify the environment, it avoids unsupported claims, and it clearly separates observed metadata from inferred risk.

Problem

- Governance reviews require users to inspect metadata across many Snowflake objects and access-control surfaces.
- Raw metadata does not easily translate into business-friendly risk findings.
- Small trial or demo environments often have incomplete governance features, so a useful skill must still return value without hallucinating results.
- Security-sensitive workflows require strong guardrails to prevent accidental changes.

Solution

- Created a read-only skill named data-governance-health-check.
- Used metadata-driven discovery rather than hardcoded object lists.
- Checked sensitive data signals, tags, protection coverage, grants, and privilege hygiene.
- Returned a concise governance health report with findings, severity, recommendations, and limitations.

Impact

The project turns a technical Snowflake metadata review into an AI-assisted evaluation workflow that a governance or platform team can understand and act on. It is especially useful for demos, early governance reviews, and repeatable health checks.

2. Skill Design

The skill is structured around five phases: environment discovery, scope and metadata discovery, analysis, severity scoring, and report generation.

Phase	Purpose	Output
0. Environment discovery	Identify current role, database, schema, warehouse, and accessible context.	Safe starting point and visible scope.
1. Metadata discovery	Inspect accessible metadata for tables, views, tags, grants, and policy references.	Inventory of governance signals.
2. Analysis	Evaluate sensitive data visibility, protection coverage, access risk, and privilege hygiene.	Evidence-backed findings.
3. Severity scoring	Classify risks conservatively as High, Medium, or Low.	Prioritized issues.
4. Report generation	Convert findings into a readable governance health report.	Executive summary, recommendations, and limitations.

Key design choices

- Read-only guardrails: the skill avoids create, alter, drop, grant, revoke, update, and delete actions.
- Metadata-driven: it discovers scope and governance signals from Snowflake metadata rather than relying on fixed table names.
- Cautious risk language: it says "no visible masking controls were found" instead of claiming confirmed exposure without evidence.
- Graceful degradation: unsupported or unavailable metadata becomes a limitation, not a failed workflow.
- Business-readable output: the final report is structured for both technical and non-technical stakeholders.

3. Core Skill Instructions

The SKILL.md file defines when to use the workflow, what metadata to inspect, how to assign severity, and how the final report should be written.

Core purpose

Use this skill to perform a read-only governance posture review of a Snowflake environment. The skill inspects accessible metadata to identify signs of sensitive data, governance tagging coverage, missing or unvalidated protection controls, and broad or risky access patterns.

Scope of review

- Sensitive-looking columns and governed datasets
- Tag coverage across tables and columns
- Masking policy and row access policy visibility
- Broad grants, future grants, and least-privilege review candidates
- Metadata limitations and unsupported features

Guardrails

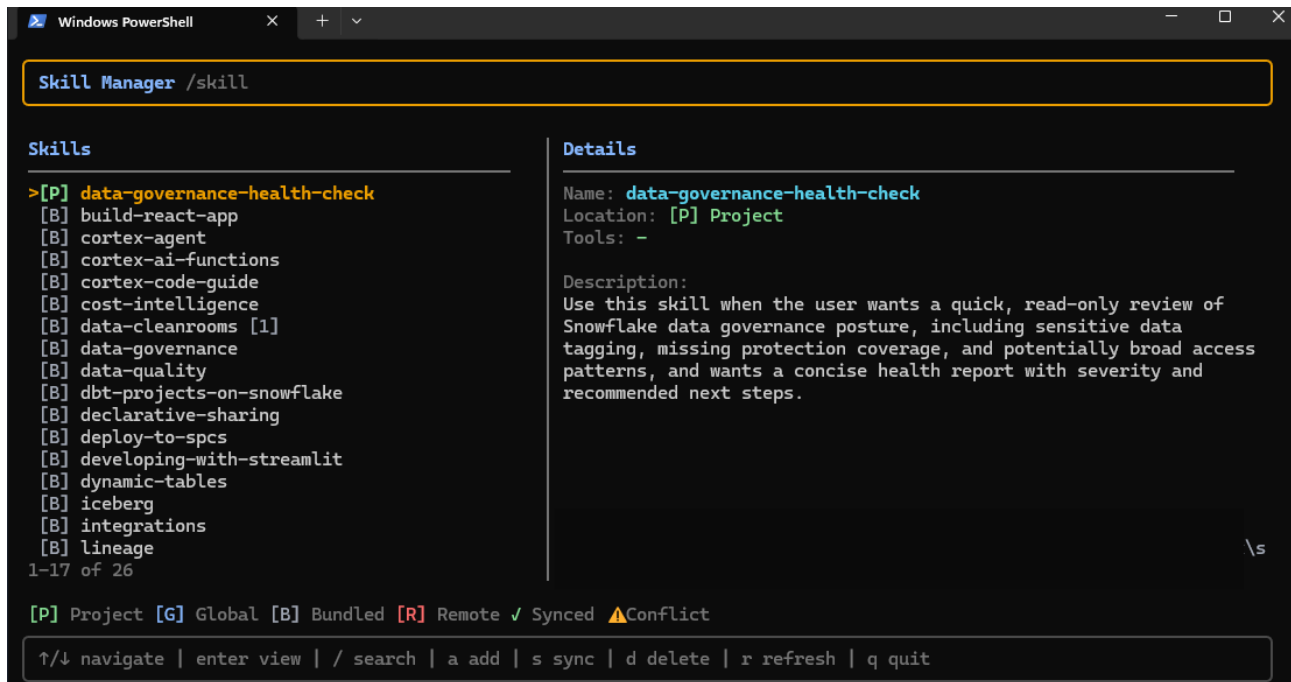
The workflow must remain safe and non-destructive. It should never fabricate governance controls, overstate certainty, or fail completely when optional metadata is unavailable.

Recommended output format

Governance Health Report: executive summary, scope reviewed, findings by category, severity, recommended next steps, and limitations.

4. Implementation Screenshots

The skill was loaded and tested in Cortex Code CLI against a demo Snowflake governance environment. Sensitive personal and account-specific details have been removed from screenshots for public sharing.



```
Windows PowerShell
Skill Manager /skill

Skills
>[P] data-governance-health-check
[B] build-react-app
[B] cortex-agent
[B] cortex-ai-functions
[B] cortex-code-guide
[B] cost-intelligence
[B] data-cleanrooms [1]
[B] data-governance
[B] data-quality
[B] dbt-projects-on-snowflake
[B] declarative-sharing
[B] deploy-to-spcs
[B] developing-with-streamlit
[B] dynamic-tables
[B] iceberg
[B] integrations
[B] lineage
1-17 of 26

Details
Name: data-governance-health-check
Location: [P] Project
Tools: -

Description:
Use this skill when the user wants a quick, read-only review of
Snowflake data governance posture, including sensitive data
tagging, missing protection coverage, and potentially broad access
patterns, and wants a concise health report with severity and
recommended next steps.

[P] Project [G] Global [B] Bundled [R] Remote ✓ Synced ⚠ Conflict
↑/↓ navigate | enter view | / search | a add | s sync | d delete | r refresh | q quit
```

Figure 1. Skill installed in the CLI skill manager

5. Live Run: Environment and Metadata Discovery

The run started by checking the current Snowflake context, confirming the target demo database, listing schemas, and discovering tables in scope.

```
Cortex Code v1.0.36

> Use the data-governance-health-check skill to review GOVERNANCE_DEMO and summarize sensitive data coverage, missing protection controls, and risky access patterns.

✓ SKILL data-governance-health-check
Use this skill when the user wants a quick, read-only review of Snowflake dat...

    t current Snowflake context

SQL Result (1 rows x 4 cols)


| # | CURRENT_ROLE() | CURRENT_DATABASE() | CURRENT_SCHEMA() | CURRENT_WAREHOUSE() |
|---|----------------|--------------------|------------------|---------------------|
| 1 | ACCOUNTADMIN   | None               | None             | COMPUTE_WH          |



    ack if GOVERNANCE_DEMO database exists and list schemas

Connection: 5139486
SQL Result (4 rows x 14 cols) [ctrl+t to navigate]


| # | created_on                     | name               | is_default | is_current |
|---|--------------------------------|--------------------|------------|------------|
| 1 | "2026-03-14 21:57:38.294 -0... | INFORMATION_SCHEMA | N          | N          |
| 2 | "2026-03-14 17:24:43.626 -0... | PUBLIC             | N          | N          |
| 3 | "2026-03-14 17:24:44.024 -0... | RAW                | N          | N          |



Rows 1-3/4 | Cols 1-4/14

    st tables in GOVERNANCE_DEMO

SQL Result (3 rows x 22 cols) [ctrl+t to navigate]


| # | created_on                     | name          | database_name   | schema_name | kind  |
|---|--------------------------------|---------------|-----------------|-------------|-------|
| 1 | "2026-03-14 17:25:27.667 -0... | CUSTOMERS     | GOVERNANCE_DEMO | RAW         | TABLE |
| 2 | "2026-03-14 17:25:29.479 -0... | ORDERS        | GOVERNANCE_DEMO | RAW         | TABLE |
| 3 | "2026-03-14 17:26:05.770 -0... | REGION_ACCESS | GOVERNANCE_DEMO | RAW         | TABLE |



Rows 1-3/3 | Cols 1-5/22
```

Figure 2. Metadata discovery across current context, schemas, and tables

6. Live Run: Table and Tag Checks

The skill inspected table columns and governance tags, including customer and order-related objects. This helped identify which fields were tagged and which areas had incomplete classification coverage.

Describe ORDERS table columns

SQL Result (6 rows x 12 cols) [ctrl+t to navigate]

#	name	type	kind	null?	default	primary key	unique key	check
1	ORDER_ID	NUMBER(38,0)	COLUMN	Y	None	N	N	None
2	CUSTOMER_ID	NUMBER(38,0)	COLUMN	Y	None	N	N	None
3	ORDER_STATUS	VARCHAR(1)	COLUMN	Y	None	N	N	None

Rows 1-3/6 | Cols 1-8/12

✓ SNOWFLAKE_SQL_EXECUTE List roles in account
Connection: ST39406
SQL Result (8 rows x 11 cols) [ctrl+t to navigate]

		name	is_default	is_current
1	"2026-03-14 14:31:24.251 -0...	ACCOUNTADMIN	Y	Y
2	"2026-03-14 17:35:49.608 -0...	GOVERNANCE_DEMO_ANALYST	N	N
3	"2026-03-14 14:31:24.571 -0...	ORGADMIN	N	N

Rows 1-3/8 | Cols 1-4/11

✓ SNOWFLAKE_SQL_EXECUTE Describe CUSTOMERS table columns
Connection: ST39406
SQL Result (7 rows x 12 cols) [ctrl+t to navigate]

		type	kind	null?	default	primary key	unique key
1	CUSTOMER_ID	NUMBER(38,0)	COLUMN	Y	None	N	N
2	CUSTOMER_NAME	VARCHAR(25)	COLUMN	Y	None	N	N
3	EMAIL	VARCHAR(16777216)	COLUMN	Y	None	N	N

Rows 1-3/7 | Cols 1-7/12

✓ SNOWFLAKE_SQL_EXECUTE List all tags in account
Connection: ST39406
SQL Result (4 rows x 9 cols) [ctrl+t to navigate]

#	created_on	name	database_name
1	"2026-03-14 17:26:47.415 -0...	DATA_DOMAIN	GOVERNANCE_DEMO
2	"2026-03-14 17:26:47.120 -0...	SENSITIVITY	GOVERNANCE_DEMO
3	"2025-06-24 02:09:44.747 -0...	SKIP_SENSITIVE_DATA_CLASSIF...	SNOWFLAKE

Rows 1-3/4 | Cols 1-3/9

Figure 3. Column and tag metadata checks

7. Live Run: Policy and Grant Checks

The workflow checked visible masking policy metadata, grants to the demo analyst role, tag references, and optional row access policy metadata. Unsupported checks were captured as limitations instead of breaking the run.

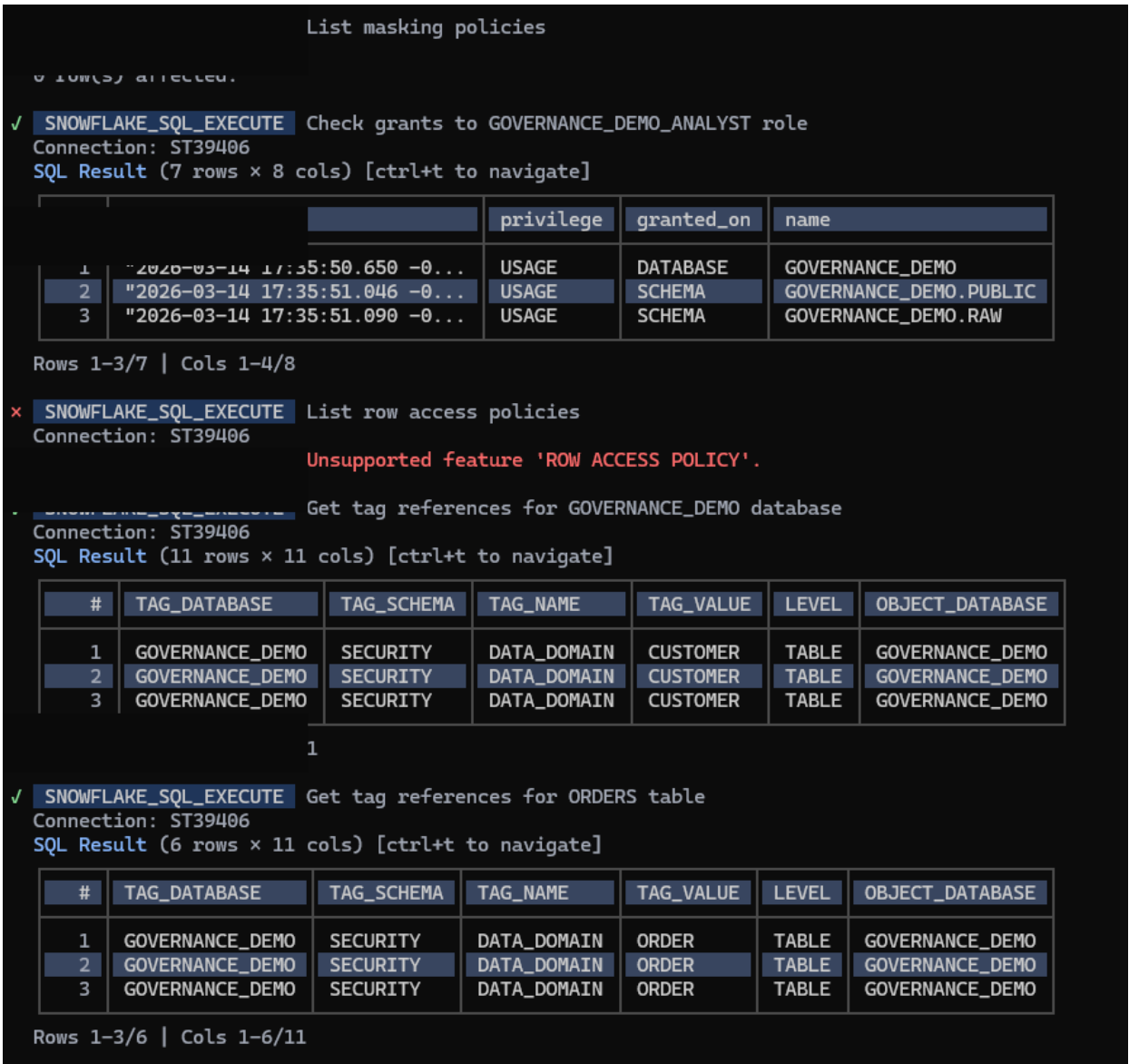


Figure 4. Policy visibility, grant review, and tag-reference checks

future grants in GOVERNANCE_DEMO

SQL Result (1 rows x 7 cols) [ctrl+t to navigate]

#	created_on	privilege	grant_on	name	grant_to
1	2024-09-01 10:00:00	SELECT	TABLE	GOVERNANCE_DEMO.<TABLE>	ROLE

Rows 1-1/1 | Cols 1-5/7

SHOWFLAKE_SQL_EXECUTE Describe REGION_ACCESS table
Connection: ST39406

) [ctrl+t to navigate]

#	name	type	kind	null?	default	primary key	unique key
1	ROLE_NAME	VARCHAR(16777216)	COLUMN	Y	None	N	N
2	REGION	VARCHAR(16777216)	COLUMN	Y	None	N	N

SHOWFLAKE_SQL_EXECUTE Check column-level policy references in account_usage
Connection: ST39406

Error executing SQL: SQL compilation error: error line 1 at position 104

New Session

Governance Health Report: GOVERNANCE_DEMO

Executive Summary

The GOVERNANCE_DEMO database has partial governance implementation. Sensitivity tags are applied to key PII columns (EMAIL, PHONE), but no masking policies exist to protect this HIGH-sensitivity data. A future grant automatically gives GOVERNANCE_DEMO_ANALYST SELECT on all new tables, creating broad access. Row access policies are not supported in this account.

Figure 5. Additional metadata checks and early report output

8. Generated Governance Report

The final output summarized the scope, tagged sensitive fields, missing or unvalidated protection coverage, and access-risk findings in a structured report.

Scope Reviewed	
Component	Details
Database	GOVERNANCE_DEMO
Schemas	RAW, SECURITY, PUBLIC
Tables	CUSTOMERS (200 rows), ORDERS (500 rows), REGION_ACCESS (4 rows)
Tags	SENSITIVITY (LOW/MEDIUM/HIGH), DATA_DOMAIN (CUSTOMER/ORDER)
Role	GOVERNANCE_DEMO_ANALYST

Findings	
1. Sensitive Data Visibility	
Severity	Medium
Observed: Governance tags are applied:	
- EMAIL → SENSITIVITY=HIGH - PHONE → SENSITIVITY=HIGH - CUSTOMER_NAME → SENSITIVITY=MEDIUM - ACCOUNT_BALANCE → SENSITIVITY=LOW - All CUSTOMERS columns → DATA_DOMAIN=CUSTOMER - All ORDERS columns → DATA_DOMAIN=ORDER	
Why it matters: Good tagging foundation exists. However, ORDERS table columns lack sensitivity tags (only domain tags).	
Recommendation: Expand SENSITIVITY tagging to ORDERS.TOTAL_PRICE (financial data).	

2. Protection Coverage	
Severity	HIGH

Figure 6. Scope reviewed and sensitive data visibility findings

9. Risk Summary and Recommendations

The report ended with a risk summary, limitations, and prioritized next steps. The highest-priority issue was to add protection controls before broad analyst access is granted.

Observed:

- GOVERNANCE_DEMO_ANALYST has SELECT on all 3 tables (CUSTOMERS, ORDERS, REGION_ACCESS)
- Future grant exists: Auto-grants SELECT on all new tables in GOVERNANCE_DEMO to this role
- HIGH-sensitivity data (EMAIL, PHONE) is readable by GOVERNANCE_DEMO_ANALYST with no masking

Why it matters: Future grants automatically expand access to new sensitive data. Combined with missing masking, this creates cumulative risk.

Recommendation:

1. Remove or scope down the future grant

2. Apply masking policies before granting access to sensitive columns

3. Consider role-based masking tied to REGION_ACCESS table

4. Privilege Hygiene

Severity

Low

Observed: GOVERNANCE_DEMO_ANALYST has access to all 3 schemas including SECURITY (contains governance tags). Usage metadata not available to confirm actual usage.

Recommendation: Review whether SECURITY schema access is needed for analyst role.

Summary Table

Category	Severity	Key Issue
Sensitive Data	Medium	ORDERS lacks sensitivity tags
Protection Coverage	HIGH	No masking on HIGH-sensitivity EMAIL/PHONE
Access Risk	HIGH	Future grant + unmasked PII access
Privilege Hygiene	Low	Broad schema access

Figure 7. Protection coverage, privilege hygiene, and risk summary

Limitations

Check	Status
Masking policy coverage	✔ Checked (none found)
Row access policies	✖ Unsupported in this accou...
Activity/usage metadata	✖ Not accessible
Classification profiles	✖ Not configured

Recommended Next Steps

1. Immediate: Create masking policies for EMAIL and PHONE columns

2. Short-term: Review/remove future grant on GOVERNANCE_DEMO tables

3. Medium-term: Add SENSITIVITY tags to financial columns in ORDERS

4. Consider: Evaluate upgrading to an edition that supports row access policies for region-based filtering

Governance health check complete. Key findings:

- 2 HIGH-severity issues: (1) No masking policies protect HIGH-sensitivity EMAIL/PHONE columns, (2) Future grant auto-expands access to GOVERNANCE_DEMO_ANALYST
- Tagging foundation exists but ORDERS table lacks sensitivity classifications
- Row access policies unsupported in this account edition

Top priority: Create masking policies for EMAIL and PHONE before broader analyst access is granted.

Figure 8. Limitations and recommended next steps

10. Evaluation Plan

The evaluation focused on correctness, reliability, clarity, and graceful handling of incomplete or unsupported governance features.

Test	Category	Scenario	Expected behavior
T1	Happy path	Demo database exists with tables, tags, grants, and visible metadata.	Discover scope, inspect metadata, and generate a governance report.
T2	Happy path	User asks for narrower grant and least-privilege review.	Prioritize roles, grants, future grants, and access-risk findings.
T3	Edge case	Sensitive fields exist but masking policies are absent or not visible.	Flag missing or unvalidated protection coverage and continue normally.
T4	Edge case	Small demo environment with limited objects.	Return a useful scope-aware review without overstating conclusions.
T5	Failure handling	Optional row access policy check is unsupported.	Continue the run and list row-level checks as a limitation.
T6	Failure handling	One optional metadata query returns a SQL compilation error.	Finish remaining checks and include incomplete coverage in limitations.

Regression signals to monitor

- Increase in SQL execution errors or unsupported feature failures.
- Incorrect severity scorings or recommendations not aligned to visible metadata.
- Reduced clarity or inconsistency in the final report.
- Failure to explain limitations when metadata is incomplete.

11. Observed Results

The skill was manually tested in Cortex Code CLI against a real Snowflake trial environment using a demo governance database.

Run	Outcome
Scoped governance review	Successfully discovered the demo database, listed schemas and tables, checked tags, inspected grants, and generated a structured governance health report.
Rerun after skill refinement	Returned the same core findings with clearer wording, stronger guardrails, and more cautious handling of incomplete metadata.
Unsupported row access policy check	The workflow continued instead of failing and recorded the unsupported feature in the limitations section.
Single metadata query error	The workflow completed the remaining checks and produced the report, while noting incomplete coverage.

Key findings from the demo run

- Governance tags were applied to some sensitive fields, including customer contact data.
- Some order-related fields lacked complete sensitivity tagging.
- No visible masking policies were found for high-sensitivity fields in the demo environment.
- A demo analyst role had broad access patterns that merit least-privilege review.
- Unsupported or unavailable metadata checks were surfaced as limitations rather than hidden.

12. Product Reflection

This project is both a technical workflow and a product-design exercise. The value comes from making a complex governance review feel safe, guided, and useful for enterprise users.

What worked well

- The skill produced useful findings from live Snowflake metadata.
- The output was structured around risk, impact, and next steps rather than raw SQL results.
- The workflow handled unsupported features and query errors without collapsing.
- The read-only design made the skill safer for trial and production-like environments.

What I would improve next

- Add stronger defensive checks before running edition-specific metadata queries.
- Make the workflow even more scope-first so it stays tightly focused on the requested database or schema.
- Add a dashboard view that summarizes risk by schema, table, and severity.
- Add scheduled health checks and integration with alerts or tasks.
- Create a packaged demo environment and sample prompts for faster onboarding.